Quasi-Z-Source Inverter for Photovoltaic Power Generation Systems

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***Abstract*-This paper presents a quasi-Z-source inverter (qZSI) that is a new topology derived from the traditional Z-source inverter (ZSI). The qZSI inherits all the advantages of the ZSI, which can realize buck/boost, inversion and power conditioning in a single stage with improved reliability. In addition, the proposed qZSI has the unique advantages of lower component ratings and constant dc current from the source. All of the boost control methods that have been developed for the ZSI can be used by the qZSI. The qZSI features a wide range of voltage gain which is suitable for applications in photovoltaic (PV) systems, due to the fact that the PV cell's output varies widely with temperature and solar irradiation. Theoretical analysis of voltage boost, control methods and a system design guide for the qZSI in PV systems are investigated in this paper. A prototype has been built in the laboratory. Both simulations and experiments are presented to verify the proposed concept and theoretical analysis.**

I. INTRODUCTION

Photovoltaic (PV) power generation is becoming more promising since the introduction of the thin film PV technology due to its lower cost, excellent high temperature performance, low weight, flexibility, and glass-free easy installation. However, there are still two primary factors limiting the widespread application of PV power systems. The first is the cost of the solar cell/module and the interface converter system; the second is the variability of the output (diurnal and seasonal) of the PV cells. A PV cell's voltage varies widely with temperature and irradiation, but the traditional voltage source inverter (VSI) cannot deal with this wide range without over-rating of the inverter, because the VSI is a buck converter whose input dc voltage must be greater than the peak ac output voltage. Because of this. a transformer and/or a dc/dc converter is usually used in PV applications, in order to cope with the range of the PV voltage , reduce inverter ratings, and produce a desired voltage for the load or connection to the utility. This leads to a higher component count and low efficiency, which opposes the goal of cost reduction.

The Z-source inverter (ZSI) has been reported suitable for residential PV system [1-2] because of the capability of voltage boost and inversion in a single stage. Recently, four new topologies, the quasi-Z-source inverters (qZSI), have been derived from the original ZSI [3]. This paper analyzes one voltage fed topology of these four in detail and applies it to PV power generation systems. By using the new quasi-Z-source topology, the inverter draws a constant current from the PV array and is capable of handling a wide input voltage range. It also features lower component ratings and reduced source stress compared to the traditional ZSI. A prototype which provides three phase 60-Hz, 208-V_{llrms} ac has been built in laboratory. It is demonstrated from the theoretical analysis, simulation and experimental results that the proposed qZSI can realize voltage buck or boost and dc-ac inversion in a single stage with high reliability and efficiency, which makes it well suited for PV power systems.

II. CIRCUIT ANALYSIS OF THE QUASI-Z-SOURCE INVERTER

A. *Quasi-Z-Source Inverter Circuit*

Figs. 1a and 1b show the traditional voltage fed ZSI [4] and the proposed voltage fed qZSI, respectively. In the same manner as the traditional ZSI, the qZSI has two types of operational states at the dc side: the nonshoot-through states (i.e. the six active states and two conventional zero states of the traditional VSI) and the shoot-through state (i.e. both switches in at least one phase conduct simultaneously).

In the non-shoot-through states, the inverter bridge viewed from the dc side is equivalent to a current source. The equivalent circuits of the two states are as shown in Figs. 2a and 2b. The shoot-through state is forbidden in the traditional VSI, because it will cause a short circuit of the voltage source and damage the devices. With the qZSI and ZSI, the unique LC and diode network connected to the inverter bridge modify the operation of the circuit, allowing the shoot-through state. This network will effectively protect the circuit from damage when the shoot-through occurs and by using the shoot-though state, the (quasi-) Z-source network boosts the dc-link voltage. The major differences between the ZSI and qZSI are (1) the qZSI draws a continuous constant dc current from the source

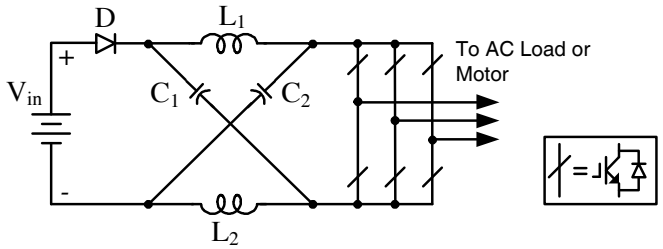


Figure 1a. Voltage fed Z-source inverter

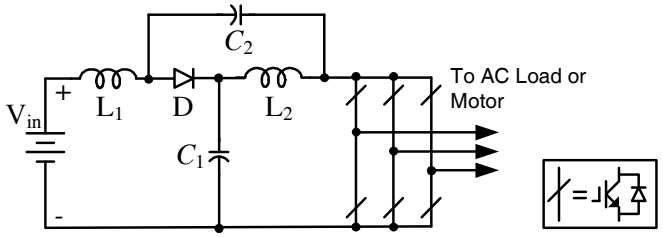


Figure 1b. Voltage fed quasi-Z-source inverter

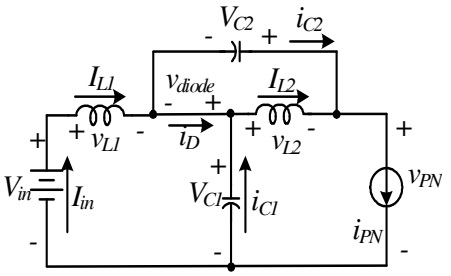


Figure 2a. Equivalent circuit of the qZSI in non-shoot-through states

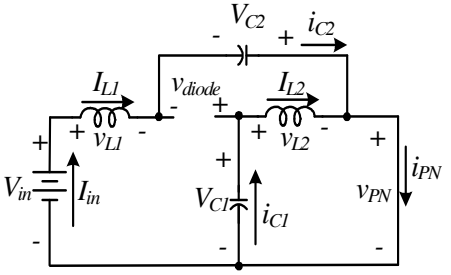


Figure 2b. Equivalent circuit of the qZSI in shoot-through states

while the ZSI draws a discontinuous current and (2) the voltage on capacitor C_2 is greatly reduced. The continuous and constant dc current drawn from the source with this qZSI make this system especially well suited for PV power conditioning systems.

B. Circuit Analysis

All the voltages as well as the currents are defined in Figs 2a, 2b and the polarities are shown with arrows. Assuming that during one switching cycle, T, the interval of the shoot-through state is T_0; the interval of non-shoot-through states is T_1; thus one has $T = T_0 + T_1$ and the shoot-through duty ratio, $D = T_0 / T$. From Fig 2a which is a representation of the inverter during the interval of the non-shoot-through states, T_1, one can get

$$v_{L1} = V_{in} - V_{C1}, \qquad v_{L2} = -V_{C2}, \text{ and} \tag{1}$$

$$v_{PN} = V_{C1} - v_{L2} = V_{C1} + V_{C2} \quad v_{diode} = 0. \tag{2}$$

From Fig 2b which is a representation of the system during the interval of the shoot-through states, T_0, one can get

$$v_{L1} = V_{C2} + V_{in}, \qquad v_{L2} = V_{C1}, \text{ and} \tag{3}$$

$$v_{PN} = 0 \quad v_{diode} = V_{C1} + V_{C2}. \tag{4}$$

At steady state, the average voltage of the inductors over one switching cycle is zero. From (1), (3), one has

$$\begin{cases} V_{L1} = \overline{v}_{L1} = \dfrac{T_0(V_{C2} + V_{in}) + T_1(V_{in} - V_{C1})}{T} = 0 \\ V_{L2} = \overline{v}_{L2} = \dfrac{T_0(V_{C1}) + T_1(-V_{C2})}{T} = 0 \end{cases}$$

Thus

$$V_{C1} = \frac{T_1}{T_1 - T_0} V_{in} \qquad V_{C2} = \frac{T_0}{T_1 - T_0} V_{in} \tag{5}$$

From (2), (4) and (5), the peak dc-link voltage across the inverter bridge is

$$\hat{v}_{PN} = V_{C1} + V_{C2} = \frac{T}{T_1 - T_0} V_{in} = \frac{1}{1 - 2\dfrac{T_0}{T}} V_{in} = B V_{in} \tag{6}$$

where B is the boost factor of the qZSI. This is also the peak voltage across the diode.

The average current of the inductors L_1, L_2 can be calculated by the system power rating P

$$I_{L1} = I_{L2} = I_{in} = P / V_{in} \tag{7}$$

According to Kirchhoff's current law and (7), we also can get that

$$I_{C1} = I_{C2} = I_{PN} - I_{L1} \qquad I_D = 2I_{L1} - I_{PN} \tag{8}$$

In summary, the voltage and current stress of the qZSI are shown in Table 1. The stress on the ZSI is shown as well for comparison, where

(1) M is the modulation index; $\hat{v}_{\ln}$ is the ac peak phase voltage; P is the system power rating;

(2) $m = T_1/(T_1 - T_0)$; $n = T_0/(T_1 - T_0)$; thus $m > 1; m - n = 1$;

(3) $B = T/(T_1 - T_0)$, thus $m + n = B$, $1 < m < B$.

From Table 1 we can find that the qZSI inherits all the advantages of the ZSI. It can buck or boost a voltage with a given boost factor. It is able to handle a shoot through state, and therefore it is more reliable than the traditional VSI. It is unnecessary to add a dead band into control schemes, which reduces the output distortion. In addition, there are some unique merits of the qZSI when compared to the ZSI:

Table 1 Voltage and average current of the qZSI and ZSI network

	$v_{L1} = v_{L2}$		v_{PN}		v_{diode}	
	T_0	T_1	T_0	T_1	T_0	T_1
ZSI	mV_{in}	$-nV_{in}$	0	BV_{in}	BV_{in}	0
qZSI	mV_{in}	$-nV_{in}$	0	BV_{in}	BV_{in}	0
	V_{C1}		V_{C2}		$\hat{v}_{\ln}$	
ZSI	mV_{in}		mV_{in}		$MBV_{in}/2$	
qZSI	mV_{in}		nV_{in}		$MBV_{in}/2$	
	$I_{in} = I_{L1} = I_{L2}$		$I_{C1} = I_{C2}$		I_D	
ZSI	P/V_{in}		$I_{PN} - I_{L1}$		$2I_{L1} - I_{PN}$	
qZSI	P/V_{in}		$I_{PN} - I_{L1}$		$2I_{L1} - I_{PN}$	

(1) The two capacitors in ZSI sustain the same high voltage; while the voltage on capacitor C_2 in qZSI is lower, which requires lower capacitor rating;

(2) The ZSI has discontinuous input current in the boost mode; while the input current of the qZSI is continuous due to the input inductor L_1, which will significantly reduce input stress;

(3) For the qZSI, there is a common dc rail between the source and inverter, which is easier to assemble and causes less EMI problems.

III. Control Methods

A. *Buck/Boost Conversion Mode*

If the inverter is operated entirely in the non-shoot-through states (Fig. 2a) the diode will conduct and the voltage on capacitor C_1 will be equal to the input voltage while the voltage on capacitor C_2 will be zero. Therefore, $v_{PN} = V_{in}$ and the qZSI acts as a traditional VSI:

$$\hat{v}_{\ln} = \frac{\hat{v}_{PN}}{2} \cdot M = \frac{V_{in}}{2} \cdot M \tag{9}$$

For SPWM $0 \le M \le 1$; and for SVPWM $0 \le M \le 2/\sqrt{3}$. Thus when D = 0, $\hat{v}_{\ln}$ is always less than $V_{in}/\sqrt{3}$ and this is called the buck conversion mode of the qZSI.

By keeping the six active states unchanged and replacing part or all of the two conventional zero states with shoot-through states, one can boost $\hat{v}_{PN}$ by a factor of B, the value of which is related to the shoot-through duty ratio, as shown in (6). This is called the boost conversion mode of the qZSI. The peak ac voltage becomes

$$\hat{v}_{\ln} = \frac{\hat{v}_{PN}}{2} \cdot M = \frac{V_{in}}{2} \cdot BM \tag{10}$$

B. *Boost Control Methods*

All the boost control methods that have been explored for the traditional ZSI (i.e. simple boost, maximum boost, maximum constant boost) [5-7] can be utilized for qZSI control in the same manner. Generally speaking, the voltage gain of the qZSI is $G = \hat{v}_{\ln}/0.5\hat{v}_{PN} = MB$, whereas the voltage stress across the inverter bridge is BV_{in}. In order to maximize the voltage gain and minimize the voltage stress on the inverter bridge, one needs to decrease the boost factor B and increase the modulation index M as much as possible.

Fig. 3 shows the voltage gain versus the modulation index of these three boost control methods. All have significantly higher gain than traditional PWM methods. Among these three boost control methods, the maximum boost control makes the most use of the conventional zero states, so it has the maximum M and the minimum voltage stress across the inverter bridge with the same voltage gain. However, it has the drawback of low-frequency ripples on the passive components

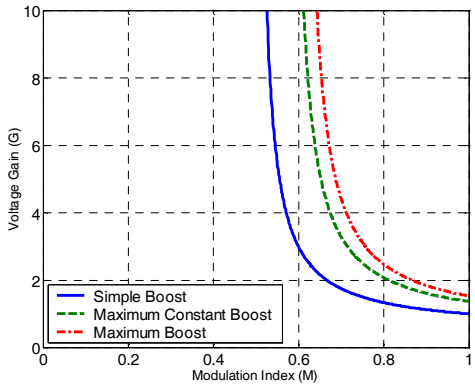


Figure 3. Voltage gain versus modulation index

of the qZSI, which requires a larger volume and weight and higher cost inductor and capacitor in the qZSI network. The simple boost control has evenly spread shoot-through states, thus it doesn't involve low-frequency ripples associated with output frequency; but its voltage stress is the largest with a given G. The maximum constant boost control makes a compromise of the two mentioned boost control methods.

In the proposed PV power generation system, in order to lower the voltage stress on the inverter bridge and keep a high voltage gain, the maximum constant boost control with third-harmonic injection was chosen as the control method. Fig. 4 shows the sketch map. At (1/6) third harmonic injection, the maximum modulation index $M = (2/\sqrt{3})$ can be achieved.

The shoot-through states are introduced into the switching cycle when the carrier is either greater than V_P or less than V_N, which is evenly spread in each switching cycle. Thus the qZSI network doesn't involve low-frequency ripples. In this case, the shoot-through duty ratio is

$$D = \frac{T_0}{T} = 1 - \frac{\sqrt{3}M}{2} \tag{11}$$

The boost factor is

$$B = \frac{1}{1-2D} = \frac{1}{\sqrt{3}M - 1} \tag{12}$$

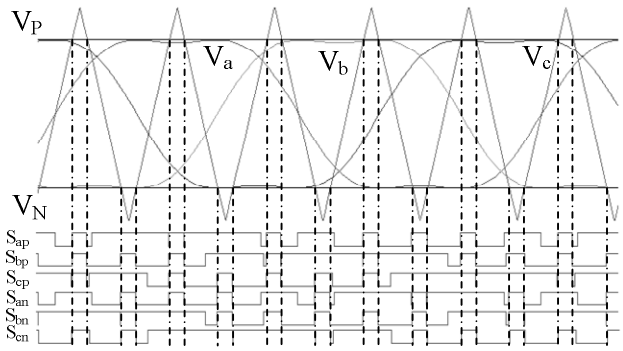


Figure 4. Sketch map of constant boost control for qZSI

And the voltage gain equals

$$G = MB = \frac{M}{\sqrt{3}M - 1} \tag{13}$$

The peak ac phase voltage can be calculated as

$$\hat{v}_{\ln} = \frac{V_{in}}{2} G = \frac{MV_{in}}{2\sqrt{3}M - 2} \tag{14}$$

IV. QZSI Design for PV Power Generation Systems

A. Voltage and Current Rating for Switches

Fig. 5 shows the proposed qZSI in the PV power generation system. It connects the PV arrays and outputs three phase 60 Hz, 208 V_{llrms} ac to resistive loads, which is the standard utility level in the United States. A three-phase LC Filter connected in Y is set right after the inverter bridge to get 60-Hz sinusoidal ac outputs.

As shown in Fig. 6, the PV arrays (VLX-80, 7×19) have nonlinear I-V characteristics, whose output voltage varies widely with temperature and irradiation. If the PV voltage varies at a 1:2 range of 200 V to 400 V, in order to output 208 V_{llrms} ac voltage, a transformer at line frequency or a dc/dc boost converter is commonly used in the traditional PV system.

However, with the boost capability, the PV power generation system based on qZSI is much simplified. According to the analysis presented earlier, if working in buck conversion mode, the minimum input voltage required by the qZSI is $V_{in} = 2\hat{v}_{\ln}/M = 294$ V (with $M = 2/\sqrt{3}$). Once the input voltage is below 294 V, the qZSI is designed to work in boost conversion mode.

Given an input PV voltage range of 200 V to 400 V, the maximum required voltage gain of the qZSI can be determined by

$$G_{\max} = \frac{\hat{v}_{\ln}}{V_{in}/2} \approx 1.7 \tag{15}$$

With maximum constant boost control method, the potential minimum modulation index M and maximum boost factor

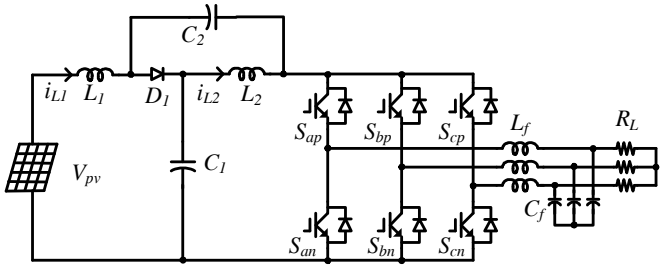


Figure 5. QZSI in the PV power generation system

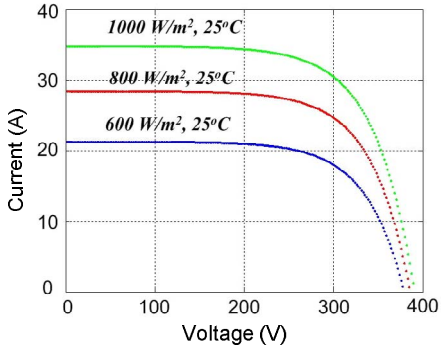


Figure 6. I-V characteristics of PV array (VLX-80)

B can be determined by (12), (13)

$$M_{\min} = \frac{G_{\max}}{\sqrt{3}G_{\max} - 1} \approx 0.875 \tag{16}$$

$$B_{\max} = \frac{1}{\sqrt{3}M_{\min} - 1} \approx 1.94 \tag{17}$$

The maximum voltage stress on the inverter bridge is predicted by $\hat{v}_{PN} = B_{\max} V_{in} \approx 388$ V. As a result, 600 V IGBTs and a 600 V diode are chosen for the proposed qZSI. According to Table 1, the voltage ratings of capacitors C_1 and C_2 are approximately 300 V and 100 V. respectively. Furthermore, with a given system power rating, e.g. 10 kW, the maximum current flow through the inductor is

$$I_{in} = I_{L1} = I_{L2} = P / V_{in} = 50 \text{ A} \tag{18}$$

B. Inductor and Capacitor Selection

In the proposed PV power generation system, the frequency of the carrier f_c is 10 kHz, and the shoot-though frequency f_s as seen by the qZSI network is doubled to 20 kHz (see Fig. 4). When the system is operating in boost conversion mode, the potential maximum interval of the shoot-though T_0, per switching cycle, can be calculated by

$$T_{0_\max} = \frac{2 - \sqrt{3}M_{\min}}{f_s} \approx 24\mu s \tag{19}$$

The inductors in the qZSI network will limit the current ripple through the devices during boost conversion mode. During shoot-though, the inductor current increases linearly. With the maximum constant boost control mode, the shoot-through interval, T_0, is evenly split into two intervals of half the duration. Choosing an acceptable peak to peak current ripple, $r_c\%$, e.g. 20% in this application, the inductance can be calculated by

$$L_1 = L_2 = \frac{V_L \Delta T}{\Delta I} = \frac{mV_{in}}{I_{L_\max} r_c\%} \cdot \frac{1}{2} T_{0_\max} \approx 356\mu H \tag{20}$$

Coupled inductors are used in this application in order to minimize the size and weight. With identical current flow, the flux is doubled for each inductor.

$$\phi = 2Ni / \Re_g = 2Ni \cdot A_L \tag{21}$$

Thus the inductance for each inductor is

$$L_1 = L_2 = N\phi / i = 2N^2 \cdot A_L \tag{22}$$

The AMCC-250 core was selected, whose A_L -Value ($\mu H / N^2$) is 0.55 when the air gap $\ell_g = 2$ *mm*. So using (22), each inductor is designed with 17 turns. The saturation current is approximately 65 A by referencing the appropriate tables on the datasheet for the AMCC-250 core.

The two capacitors are in series in the qZSI network when in the non-shoot-through states. These two capacitors absorb the current ripple and limit the voltage ripple on the inverter bridge so as to keep the output voltage sinusoidal. Assuming that the capacitance should be the same for each capacitor, the capacitance needed to limit the PN voltage ripple by $r_v\%$, e.g. 1%, can be calculated by

$$C_1 = C_2 = 2 \cdot \frac{I_C \Delta T}{\Delta(V_{C1} + V_{C2})} = 2 \cdot \frac{I_L}{BV_{in} r_v \%} \cdot \frac{1}{2} T_{0_\max} \approx 310\mu F \tag{23}$$

V. Simulation and Experimental Results

A prototype of the qZSI for PV power generation systems has been built in the laboratory to verify the analysis, the topology of which is the same as shown in Fig. 5. The system parameters for both simulations and experiments are as follows:

$$L_1 = L_2 = 500\mu H \ , \ C_1 = C_2 = 400\mu F \ ,$$
$$L_f = 1mH \ , \ C_f = 100\mu F \ , \ R_L = 10\Omega$$

The output is 60 Hz, 208 V_{llrms} ac as mentioned above, thus the system power is approximately 4.3 kW. Simulations and experiments are performed in both buck conversion mode and boost conversion mode. The input voltage changes from 400 V to 189 V. The qZSI in experiment is operated with no dead band, and there are no snubbers on either the individual switches or the dc link.

In all these cases, the timescale for waveforms of the output voltage and V_{in} in simulations is 8 milliseconds per division; the timescale for waveforms of inductor current I_L and voltages V_{pn}, V_{c1}, V_{c2} in simulations is 40 microseconds per division. The timescale for waveforms of the output voltage and V_{in} in experiments is 4 milliseconds per division; the timescale for waveforms of inductor current I_L and voltages V_{pn}, V_{c1}, V_{c2} in experiments is 20 microseconds per division.

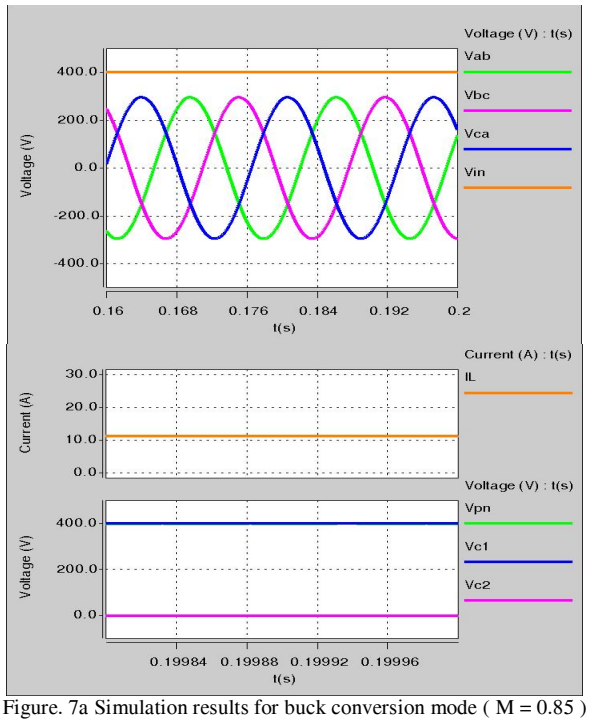


Figure. 7a Simulation results for buck conversion mode (M = 0.85)

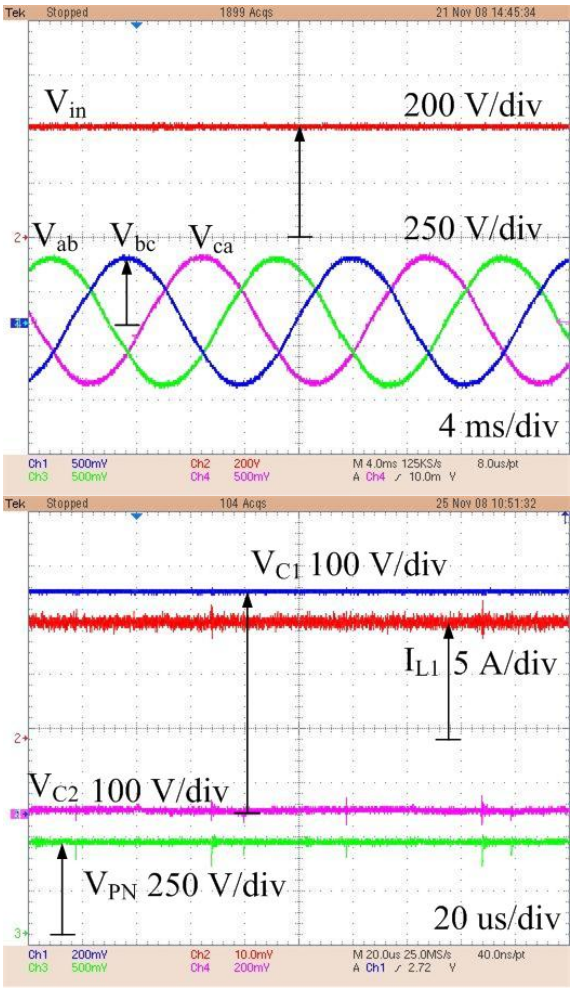


Figure. 7b Experimental waveforms for buck conversion mode (M = 0.85)

(1) Case 1: $M = 0.85; V_{in} = 400V$

Figs.7a and 7b show the simulation and experimental results in this case. The qZSI works in the buck conversion mode, operating as a conventional VSI without shoot-through states, thus $D = 0$, $B = 1$. The input voltage is 400 V; $G = M = 0.85$. The peak output line-to-line voltage was controlled to be

$$\hat{v}_{ab} = \hat{v}_{bc} = \hat{v}_{ca} = \sqrt{3} \cdot G \cdot V_{in} / 2 = 294V .$$

As can be seen in Figs. 7a and 7b, the voltage on C_1 is equal to the input voltage 400 V and the voltage on C_2 is 0 V. There is no voltage across the inductor and only a pure dc current going through the inductors. The current through L_1 measured by a current probe is 11A, which is consistent with the theoretical calculation

$$I_{L1} = P / V_{in} = 4.3k / 400 \approx 11A .$$

(2) Case 2: $M = 1; V_{in} = 249V$

Figs.8a and 8b show the simulation and experimental results in this case. As analyzed above, when V_{in} is below 294 V, the qZSI works in boost conversion mode. From (12), (13), one can get the boost factor $B \approx 1.366$ and voltage gain $G \approx 1.366$. The voltage on the dc link side will be boosted to $BV_{in} \approx 340V$; and the peak output line-to-line voltage is

$$\hat{v}_{ab} = \hat{v}_{bc} = \hat{v}_{ca} = \sqrt{3} \cdot G \cdot V_{in} / 2 \approx 294V .$$

Because there are forward voltage drop on switches and distributed resistance in the circuit, the input voltage in experiment is 254 V, which is slightly higher than the theoretical value. As can be seen in Figs. 8a and 8b, the voltage on C_1 and C_2 is 295 V and 45 V respectively, which are consistent with calculation:

$$V_{C1} = mV_{in} = 294V , \; V_{C2} = nV_{in} = 45.5V .$$

The current through L_1 measured by a current probe is 15.0 A, while the theoretical calculation is

$$I_{L1} = P / V_{in} = 4.3k / 249 \approx 17A .$$

Notice that the input current($I_{in} = I_{L1}$) is continuous, which will significantly reduce the input stress compared to the traditional ZSI.

(3) Case 3: $M = 0.85; V_{in} = 189V$

Figs.9a and 9b show the simulation and experimental results in this case. From (12) and (13), one can get the boost factor $B = 2.118$ and the voltage gain $G = 1.800$. The voltage on the dc link side will be boosted to $BV_{in} \approx 400V$; and the peak output line-to-line voltage is

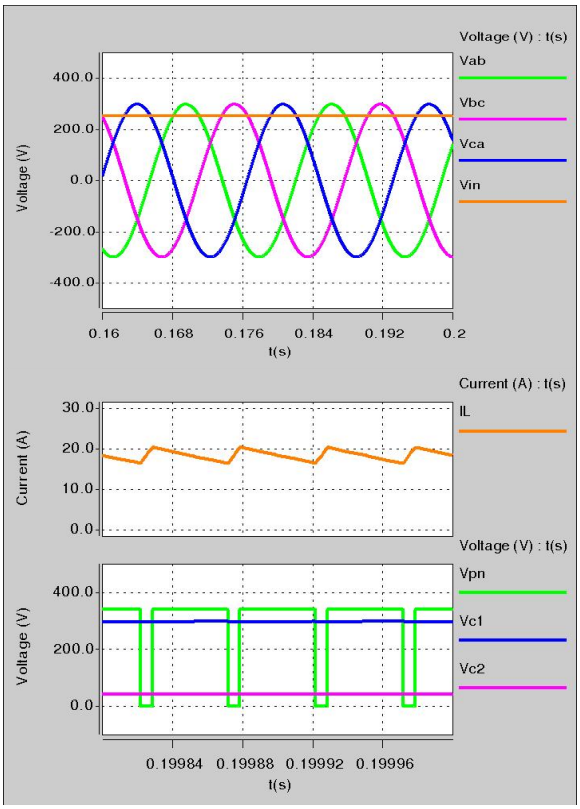


Figure. 8a Simulation results for boost conversion mode (M = 1)

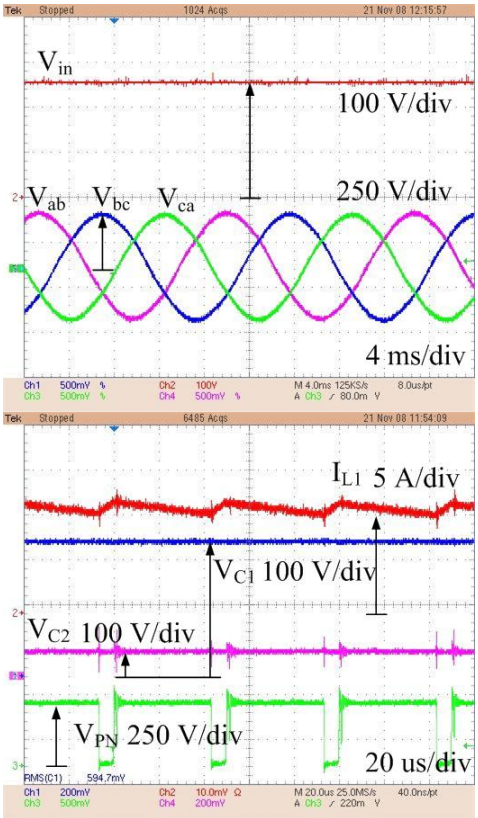


Figure. 8b Experimental waveforms for boost conversion mode (M = 1)

$$\hat{v}_{ab} = \hat{v}_{bc} = \hat{v}_{ca} = \sqrt{3} \cdot G \cdot V_{in} / 2 = 294V .$$

For the same reason, the input voltage in experiment is 195 V, which is slightly higher than the theoretical value. As can be seen in Figs. 9a and 9b, the voltage on C_1 and C_2 is 295 V and 105 V respectively, which are consistent with calculation:

$$V_{C1} = mV_{in} = 294V \ , \ V_{C2} = nV_{in} = 105.4V .$$

The current through L_1 measured by a current probe is 20.0 A, while the theoretical calculation is

$$I_{L1} = P / V_{in} = 4.3k / 189 \approx 23A .$$

VI. Conclusion and Future Work

This paper presents a quasi-Z-source inverter with a new topology, which is derived from the traditional ZSI. The proposed qZSI inherits all the advantages of the ZSI and features its unique merits. It can realize buck/boost power conversion in a single stage with a wide range of gain that is suited well for application in PV power generation systems. Furthermore, the proposed qZSI has advantages of continuous input current, reduced source stress, and lower component ratings when compared to the traditional ZSI. Theoretical analysis, control method, and system design guide are presented in this paper. Both simulation and experimental results show that with a voltage range of 1:2 at the PV input (from 189 V to 400 V), the qZSI can provide three phase 60 Hz, 208 V_{llrms} ac voltage, which verifies the theoretical analysis.

A grid-connected PV power generation system is one of the most promising applications of renewable energy sources. The proposed qZSI based PV power generation system is intended as a grid connected system and transfers the maximum power from the PV array to the grid by maximum power point tracking technology. In that case, the efficiency would be improved and the cost would be reduced with the proposed one stage power conversion system.

References

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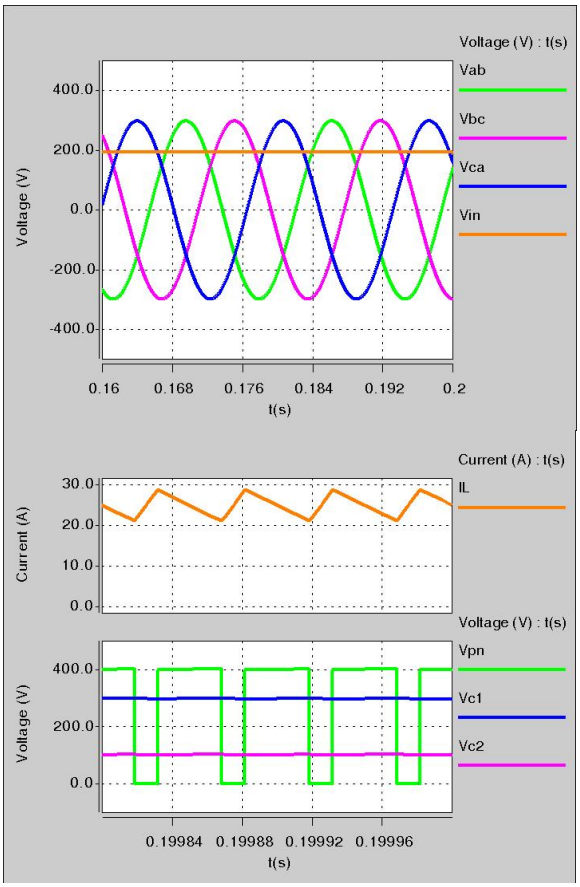


Figure. 9a Simulation results for boost conversion mode (M = 0.85)

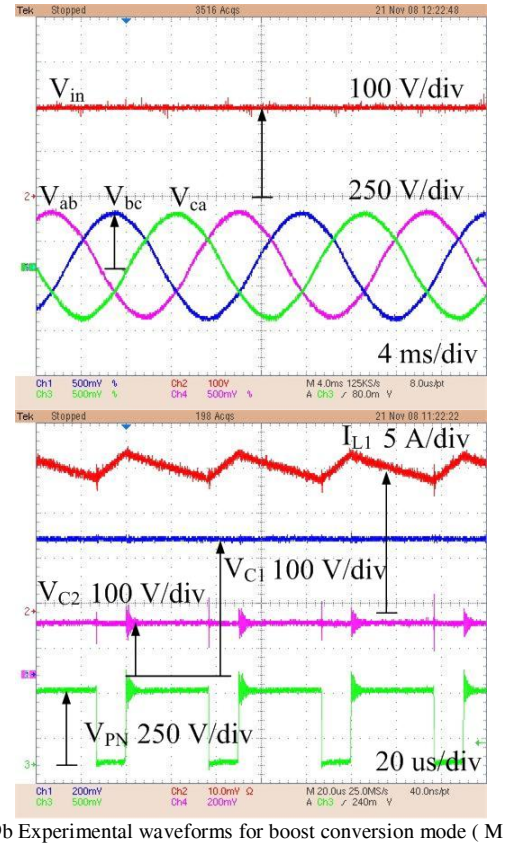


Figure. 9b Experimental waveforms for boost conversion mode (M = 0.85)